

Project Proposal

Group 28: Fraud Detection Using Graph Databases And Machine Learning.

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Abstract:

This abstract presents a project aimed at addressing the growing challenges in fraud detection within the context of digital finance, particularly the surge in fraudulent activities accompanying the exponential growth of digital banking transactions. In response to the limitations of traditional methods, the project leverages data mining, fuzzy logic, artificial intelligence (AI), machine learning (ML), and statistical techniques to automate the identification of fraudulent activities. The project employs machine learning algorithms, graph databases like Neo4j, and dimensionality reduction techniques to enhance model interpretability and performance.

The research utilizes the BankSim Dataset, a synthetic representation of mobile money transactions and digital banking that integrates both typical and fraudulent transactions. By exploring graph neural networks, Bayesian networks, support vector machines, and various classifiers, the project intends to develop an effective fraud detection model. This project holds significant potential for enhancing fraud detection methods within the digital finance landscape, with practical applications in the real-time processing of large interconnected networks.

- **Keywords:** *Machine Learning, Support Vector Machines (SVM), Random Forest Classifier, Fraud detection, Graph Convolutional Neural Network (GCN), Data Mining, Graph network, Graph Databases, Neo4j, Artificial Intelligence (AI), Synthetic data, Model evaluation, Community detection, Precision, F1-score, recall, purity, XGBoost classifier, K-Means, DBScan, N-Euler, BankSim.*

Problem Definition:

The ascent of digital payments has undeniably reshaped the financial landscape, but it has also ushered in new challenges in the form of increased fraud and financial crimes. In recent times, the realm of financial crimes and fraud has witnessed significant shifts owing to the rapid advancements in digital payment systems. The exponential growth in

digital banking transactions has led to a parallel surge in fraudulent activities, presenting fresh challenges to financial institutions and consumers alike.

Conventional methods involving manual pattern analysis and rule-based engines have demonstrated diminished effectiveness in this dynamic environment. To address these issues, we have undertaken a project that harnesses the power of data mining, fuzzy logic, artificial intelligence (AI), machine learning (ML), and statistical techniques to automate and streamline the process of identifying fraudulent activities.

The project employs machine learning algorithms for fraud detection and encompasses the following steps:

1. Refining the BankSim Dataset, Extracting Feature Information, and Conducting Data Visualization.
2. Train a model utilizing supervised learning algorithm
3. Assess the model based it's performance metrics
4. Construct a graph data model using Neo4j
5. Improve model interpretability, through feature analysis using random forest classifier and through dimensionality reduction techniques like Principle Component Analysis (PCA)
6. Train the model using the new feature set after feature reduction
7. Assess the performance metrics across a range of machine learning models.
8. Apply diverse clustering techniques to produce a graph for the purpose of grouping fraudulent data.
9. Compute evaluation metrics, and validate the model.

DataSet:

We plan to leverage the BankSim Dataset for our research in fraud detection using graph databases. This valuable dataset, also known as the "Banksim" dataset, was specifically created for this purpose. It provides a synthetic representation of mobile money transactions and digital banking, making it a crucial resource for developing and testing fraud detection models and algorithms. The dataset comprises 594,643 records, including 7,200 fraudulent transactions and 587,443 normal payments along with features such as Step, Customer Id, Age, Gender, Zip code of origin, Merchant Id, Zip Code of merchant, transaction amount and whether it is a fraudulent transaction or not. What makes it particularly useful is that it integrates both typical transactions and deliberately injected fraudulent signatures, allowing us to simulate relevant scenarios for model development. Notably, this dataset doesn't contain any personal or sensitive customer information, ensuring its suitability for academic research and development. The BankSim Dataset employs the relationships between merchants and customers, utilizing statistical and Social Network Analysis (SNA) built on Multi Agent-Based

Simulation (MABS) techniques to calibrate the model, making it an invaluable tool in advancing fraud detection methodologies in the realm of digital finance.

State-of-Art Methods & Algorithms:

Financial crimes and fraud have evolved in recent years as a result of the significant advancement in digital payments, particularly during and after the pandemic. In 2019, the digital payment network for credit cards alone processed 743 million transactions totaling \$187 billion. In recent years, there has been a significant surge in online transactions, particularly driven by the widespread adoption of credit card payments. According to the Global Payments Report of 2015, credit cards emerged as the most widely used payment method worldwide in 2014, surpassing alternatives like e-wallets and bank transfers [3]. However, this surge in credit card transactions has correspondingly led to an increase in security breaches, underscoring the critical need for effective fraud detection mechanisms in credit card transactions. As scammers quickly figure out how to get around these static constraints, traditional methods of fraud detection, such rule-based engines, have largely lost their effectiveness in this new context.

A study [4] addressed this challenge by employing supervised Bayesian Network Classifiers, including K2, Tree Augmented Naïve Bayes (TAN), and Naïve Bayes, logistics, and J48 classifiers for the purpose of detecting fraudulent credit card breaches. Notably, the study enhanced the accuracy of these models through data preprocessing techniques involving Principal Component Analysis (PCA), which involved feeding the preprocessed data into the classifiers.

In tandem with the advancements in fraud detection methods, fraudsters have exhibited increased sophistication, often collaborating to create fictitious identities that evade conventional detection methods. Traditional approaches have shown limitations in identifying such fraudulent activities. However, the utilization of graph database-based fraud detection methods has proven effective in uncovering fraud rings and detecting orchestrated fraudulent activities. A paper [5] elaborates on leveraging graph databases to identify and understand connections between data, demonstrating high accuracy in fraud detection compared to traditional database methods. Additionally, the paper delves into prevalent fraud patterns within various financial sectors including first-party banks, insurance, and e-commerce. The central premise of the paper is that deception often relies on layers of indirection, which can be unraveled through connected analysis [5].

Graph neural networks beat conventional neural networks in two areas: better generalizations and stronger relational thinking. Because they enable the construction

of architectures with a number of blocks and a certain level of structural configurability, GNNs provide flexible representations in terms of attributes and graph topologies [6]. Graph convolutional networks (GCNs) are used in financial crime detection to expose disguised criminals by examining dynamic characteristics of graph networks, which are critical in fraud applications. GCNs are used to detect money laundering. In this project, we intend to use more cutting-edge techniques like bayesian networks and support vector machines (SVM) because graph computing systems are under severe reaction time pressure due to the usage of large, interconnected networks and real-time processing restrictions. Graph-based techniques may also use a variety of intricate features, depending on the use case, and to reach approve/deny decisions for the transaction, these features must be computed in real-time [6].

Building upon the insights gained from these cutting-edge models and methodologies, it is plausible to apply similar approaches to the BankSim dataset. Specifically, utilizing graph databases such as Neo4j in tandem with classification algorithms like Gaussian Classifier, Support Vector Machine (SVM), or XGBoost Classifier can aid in the identification and classification of fraudulent and non-fraudulent transactions.

Research Plan:

During our analysis of the BankSim dataset for fraud detection, we discovered some potential challenges with the data distribution that could impact the validity of the results. The dataset has a highly imbalanced class distribution. For every 1 fraudulent transaction there are 100 legitimate transactions. This imbalance makes it challenging to properly train and evaluate models because the model might classify most of the transactions as legitimate transactions and will get an accuracy of ~99%. So, to counter that we can either under sample total legitimate transactions or we can increase the total number of fraudulent transactions.

One of the issues that we might face with randomly under sampling or over sampling is that in certain scenarios we might remove important/relevant transactions of a particular user which will affect its total indegree and outdegree count, and it might lead to wrong feature selection. To counter that, we are going to analyze the data and won't be pruning the data randomly. So instead, we will consider all the transactions of a specific user for training if the user is selected for under sampling data.

Following the preprocessing the data, we will be using certain classifiers as the baseline model such as SVM classifier, bayesian network, random forest classifier and neural nets model and to evaluate the performance based on the features of the dataset like Age, Amount, Time, Category etc. We will use the above mentioned classifiers to find

out the most significant features and will be computing the performance metrics like accuracy, F-1 score, Precision and recall to note most relevant features.

Also as a part of the research plan we plan loading the contents of the BankSim dataset on using the graph database Neo4j to produce the graph based features. We will be utilizing N-Euler graph based features like Pagerank, Degree, Betweenness etc. Furthermore, we plan to utilize clustering techniques like K-Means and DBScan using the above features to cluster and identify the fraudulent data and evaluate the cluster validation using purity.

Evaluation Plan :

The effectiveness of the fraud detection model is gauged using various model performance metrics and additional key aspects to ensure a well-rounded evaluation. The model's predictive results contain two categories: the true class and the predicted class, which form the basis for deriving critical performance metrics. The evaluation pivots on:

Precision: Accuracy in identifying actual fraud cases.

Recall/Sensitivity: Effectiveness in retrieving all relevant fraud instances.

Accuracy: Overall correctness in classifications.

F-Score: Balance between precision and recall.

ROC-AUC: Assessing the trade-off between true positive rate and false positive rate.

Specificity: Proportion of actual negatives that are correctly identified.

Outlier Sensitivity: Consistency in predictions with anomalous data.

Project Timeline & Division of Work:

Name	Task	Proposed Deadline
Bitaan, Aman	Acquire and analyze the BankSim Dataset, addressing class imbalance while preserving transactions.	10/13/23
Aman, Rahil	Train baseline models without graph based features on the pre-processed data. Evaluating the models and assessing the feature relations.	10/18/23
Rahil, Priya	Assessing model performance based on feature correlation, dimensionality reduction using PCA and identifying potential advanced classification models on evaluation	10/26/23

Rithik, Aniket	Loading dataset in graph DB and assessing, visualizing the relations generated.	10/30/23
Aniket, Bitaan	Extracting graph-based features like PageRank. Analyze extracted features and determining their relevance to problem statements.	11/01/23
Priya, Rithik	Train Graph ML models using the new dataset and relations. Evaluate GNN models using predefined performance metrics.	11/04/23
Aman, Rahil	Compare results with baseline models and note improvements or drawbacks.	11/04/23

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